Mapping Wildfire Vulnerability in Attica, Greece: A Spatial Assessment of Historical Fire Exposure and Emergency Service Accessibility for Strategic Infrastructure Planning

1. Introduction

On July 23, 2018, the coastal village of Mati in Attica, Greece, became the site of Europe's deadliest wildfire since 1900. Within hours, flames raced through pine forests and residential areas killing 104 people— many bodies found mere meters from the sea, having fled but unable to escape in time. Survivors described receiving no evacuation warnings, ran into blocked escape routes, and waited hours for coastguard rescue while fires consumed their neighborhoods. In the post-incident investigation, it was revealed that, not only were there extreme weather conditions, but the fundamental failures in emergency response coordination and infrastructure became evidently clear. Fire service officials were later convicted of criminal negligence for poorly coordinating the response, diverting a helicopter that should have aided firefighting efforts, and failing to recommend evacuations or warn the public. Political officials including the regional governor and local mayors were acquitted, sparking outrage among victims' families and underscoring persistent accountability gaps in Greece's wildfire management system.

The Mati tragedy exposed a critical vulnerability in Greece’s fire-prone regions: the spatial disconnect between where fires occur and where emergency response capacity exists. This governance failure resulted in an extraordinary economic cost. In 2023 alone, wildfires caused approximately €1.8 billion in damages across Greece, representing nearly half of Europe's total €4.1 billion wildfire-related losses that year. These figures identify a fundamental policy challenge where Greek wildfire spending is overwhelmingly skewed toward suppression rather than prevention. As climate projections indicate that southern and eastern Greece will experience up to forty additional days of high fire danger annually relative to the late 20th century, the need for evidence-based infrastructure planning has never been clearer.

Recent Greek policy responses demonstrate growing recognition of these challenges. The €2.1 billion AEGIS program, announced in 2024, represents Greece's largest-ever civil protection procurement initiative, funded through the EU Recovery and Resilience Facility and European Investment Bank. The program aims to acquire over 1,000 firefighting vehicles, amphibious aircraft, drones, and fire control centers. Complementing this, Greece became the first nation worldwide to implement a national satellite-based wildfire monitoring system through a €20 million partnership with OroraTech, aiming to deploy four thermal imaging satellites by 2026. These investments signal a strategic pivot toward enhanced detection and rapid response capacity. However, the spatial questions remain: *where* should this expanded infrastructure be positioned? Which communities face the greatest combined exposure to historical fire risk and limited access to emergency services? How can public funds be more strategically allocated to maximize protection for vulnerable populations?

This analysis addresses these questions through spatial assessment of wildfire vulnerability in the Attica (Capital) region, combining twenty-four years of historical fire incident data with current fire station locations and population density metrics. By integrating historical fire exposure patterns, distance analysis to nearest fire stations, and demographic data, this study identifies priority municipalities where high fire risk converges with under-served emergency access and higher population density. The findings provide a concrete spatial framework for policymakers to evaluate where additional fire stations, equipment deployments, or enhanced response protocols could yield the greatest risk reduction returns, transforming reactive suppression spending into proactive resilience investment.

2. Methodology

2.1 Data Assembly and Preprocessing

Starting with Greek municipality boundaries and population density data, which were obtained from

the Eurostat GISCO 2024 Local Administrative Units dataset, specifically the LAU_RG_01M_2024_4326.gpkg file, which includes 2024 population density per km² (POP_DENS_2024) for each administrative unit. In GIS, this layer was clipped to the Attica region using the gadm41_GRC_2 "Attica" polygon from the Database of Global Administrative Areas to create the primary working layer, LAU_Attica, representing all municipalities within the study region.

The historic wildfire incident data spanning 2000–2024 was derived from the DataForGreece and Hellenic Fire Service combined database, which aggregates fire event records including geographic coordinates, incident dates, and total burned area in hectares. The raw dataset, forest-fires-combined_greece_2023.csv, required substantial preprocessing in R to address observations with missing or zero coordinate values. I removed rows with missing coordinates and subset the data to show only incidents occurring within Attica Region administrative boundaries. The cleaned dataset was exported as forest-fires_attica_xy_clean.csv and imported into QGIS as a point layer, with each point representing a discrete wildfire incident location over the 24-year study period.

Fire station locations were extracted from the Geofabrik OpenStreetMap Points of Interest shapefile for Greece. The original layer, gis_osm_pois_free_1.shp, was filtered using the attribute query fclass = 'fire_station' to isolate only fire station features, creating the FireStations layer. This layer was subsequently clipped to the Attica region boundary to produce FireStations_Attica, ensuring the analysis is focused exclusively on emergency response infrastructure serving the study area.

2.2 Aggregating Historical Fire Exposure by Municipality

To measure wildfire exposure for each municipality, individual fire incident points were aggregated to municipality boundaries using a zonal statistics approach. The QGIS "Join attributes by location (summary)" tool was applied with LAU_Attica as the input polygon layer and ForestFires_Attica as the join point layer. The intersect option was used to identify which fire incidents fell within each municipality boundary. Summary statistics were computed on two key fields: incident ID (generating id_count, the total number of fires per municipality) and burned_total field (producing burned_total_sum, the cumulative burned area in hectares, and burned_total_mean, the average burned area per incident).

These three metrics—fire incident frequency, cumulative burned area, and mean fire severity—were combined to construct a composite fire exposure score for each municipality. The composite score incorporates both the historical frequency of fire events and the spatial extent of damage, providing a more comprehensive vulnerability measure than incident counts alone. Municipalities were then able to be classified into five graduated categories based on their composite fire exposure scores. This graduated classification allows for immediate visual identification of municipalities with disproportionately high historical fire burden, while overlaying historic fire incident points and fire station locations demonstrates the baseline spatial relationship between past fire activity and current emergency infrastructure.

2.3 Measuring Distance to Nearest Fire Station and Access Classification

Access to the fire station network was measured as straight-line distance from each municipality's center to its nearest fire station. Municipality centroids were computed from the LAU_Attica_FireExposure polygon layer using the "Centroids" geoprocessing tool, providing point locations that approximate each municipality's geographic and population center. These centroids served as origin points, while FireStations_Attica provided the destination points.

Then, the QGIS "Distance to nearest hub (points)" tool was applied with municipality centroids as origin features and fire stations as hub features. This operation computed, for each municipality, the distance (in kilometers) to the closest fire station and assigned the name of that nearest station. The resulting HubDist field was then joined back to the primary municipality polygon layer, thus creating LAU_Attica_FireExposure_Access, which

integrated both the historical fire exposure score and now the access measurements in a single analytical layer.

To turn these distance measurements into usable proximity categories, I used the field calculator to create a binary AccessClass variable through a CASE expression. Municipalities with distances exceeding 10 kilometers from their nearest station were classified as "Under-served (>10 km from nearest station)," while municipalities within 10 kilometers were labeled "Within 10 km of nearest station." The 10-kilometer threshold reflects the critical window established in wildfire suppression research: the first 10-30 minutes of initial attack are the most important. At typical emergency response speeds, 10 kilometers represents roughly this critical timeframe—the difference between crews arriving when fires are still containable versus arriving when they've already grown beyond initial attack capability. This two-category system makes it easy to visually identify which municipalities have access within this critical response window and which face delays that could replicate the response failures observed during the 2018 Mati fires.

2.4 Identifying Priority Municipalities Through Multi-Criteria Integration

The final analytical step integrated all three, fire exposure, service access, and population density to identify municipalities warranting priority investment. Using the field calculator on the LAU_Attica_FireExposure_Access layer, to create the PriorityClass field through a multi-condition CASE expression. Municipalities were flagged as priority areas if they simultaneously met the three criteria: burned_total_sum exceeded a high-exposure threshold calibrated to the upper quartile of the regional distribution, AccessClass equaled "Under-served (>10 km from nearest station)," and POP_DENS_2024 exceeded a density cut-off representing above-median population concentration. Municipalities satisfying all three conditions were assigned the label "Priority (high exposure, under-served, higher-density)," while all others were grouped as "Other municipalities."

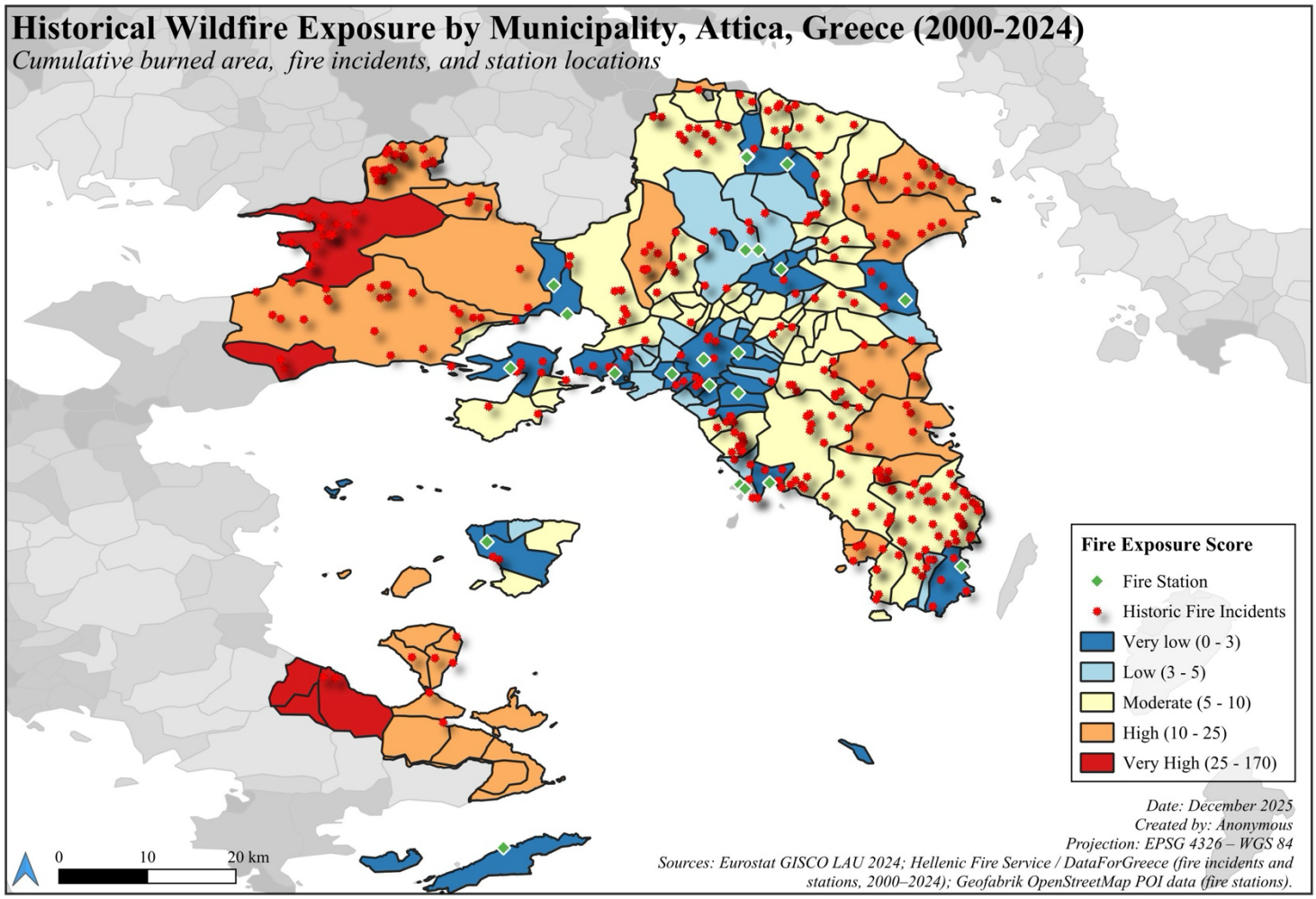
This multi-criteria approach ensures that prioritization reflects not only historical fire risk and infrastructure gaps but also the number of residents potentially affected. To determine how many people live in these priority areas, population totals were calculated into a PriorityClass category, showing what percentage of Attica's residents live in priority municipalities versus other areas. This shifts the analysis from simply identifying risky locations to counting how many people are affected, allowing policymakers to see what share of the regional population lives in places that combine high fire risk, distant emergency services, and enough residents to warrant new infrastructure investment.

2.5 Core GIS Tools and Analytical Framework

This workflow explicitly employed three key GIS tools: zonal statistics, distance/cost-distance analysis, and reclassification. Join-by-location with summary statistics functioned as a zonal statistics operation, aggregating point-level fire incidents to municipality polygons to derive exposure metrics. Distance analysis was implemented through the "Distance to nearest hub (points)" tool, computing straight-line distances from population centers to emergency infrastructure. Reclassification was executed via field calculator CASE expressions, converting continuous distance and exposure variables into categorical access classes and priority designations. Field calculator operations were particularly central in Maps 2 and 3, enabling the construction of AccessClass and PriorityClass variables and facilitating population aggregation by vulnerability category. This multi-tool integration allows the analysis to move beyond descriptive mapping of where fires occur to prescriptive identification of where people face the greatest combined exposure and access gaps, directly supporting evidence-based infrastructure planning decisions.

3. Results

3.1 Spatial Patterns of Historical Fire Exposure (Map 1)



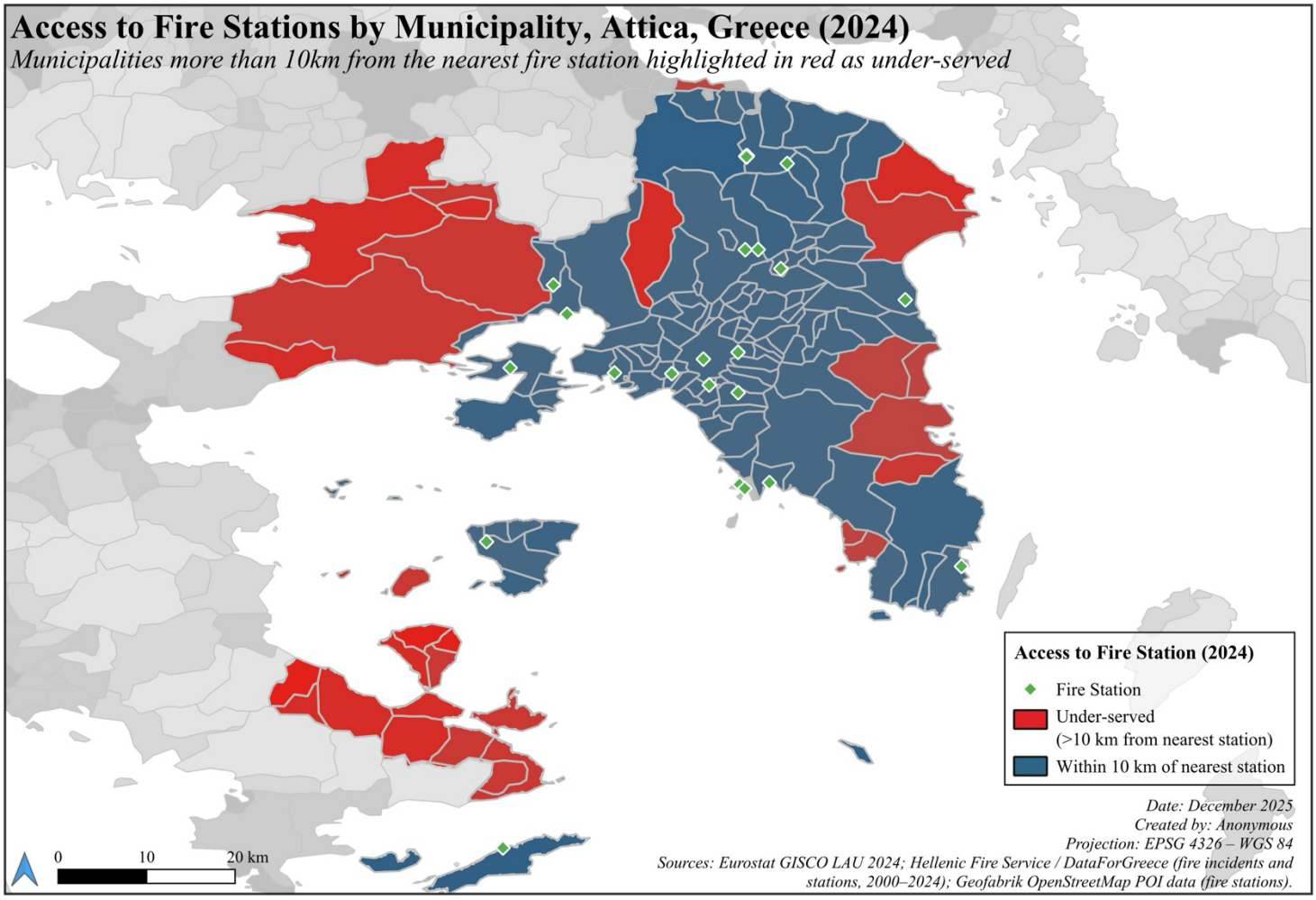
The spatial distribution of wildfire incidents across Attica from 2000 to 2024 reveals pronounced geographic clustering in the northern and eastern periphery of the Athens metropolitan area, particularly in municipalities surrounding the Penteli and Parnitha mountain ranges. The composite fire exposure classification identifies five distinct vulnerability zones based on incident frequency and cumulative burned area. Municipalities in the "Very High" exposure category, appearing in dark red, include several large administrative units in northern Attica that experienced both numerous fire events and extensive cumulative damage exceeding tens of thousands of hectares. These areas exhibit dense concentrations of red incident points, highlighting repeated fire occurrence over the 24-year study period.

In contrast, central Athens and the immediate urban core display substantially lower fire exposure, classified primarily in the "Very Low" to "Low" categories shown in dark and light blue. This urban-rural gradient reflects both reduced fuel availability in densely built environments and the concentration of wildfire risk in the wildland-urban interface where residential development meets Mediterranean vegetation. Several municipalities in the "High" and "Moderate" exposure categories, coded in orange and yellow respectively, form a transitional belt between the high-risk peripheral zones and the protected urban center, suggesting that fire

vulnerability increases with distance from the metropolitan core areas.

The overlay of current fire station locations as green diamonds demonstrates that most of the fire service infrastructure clusters in the central and coastal areas, with notably fewer stations positioned in the high-exposure northern municipalities. This spatial mismatch—where historic fire activity is most intense, but emergency infrastructure is least dense—suggests potential response time vulnerabilities that could replicate the access failures observed during the 2018 Mati fires. The graduated symbology makes clear that fire exposure is not uniformly distributed across Attica, but instead concentrates in specific municipalities that have a disproportionate historical fire burden, raising questions about whether current infrastructure placement adequately serves these repeatedly impacted communities.

3.2 Emergency Service Access Disparities (Map 2)

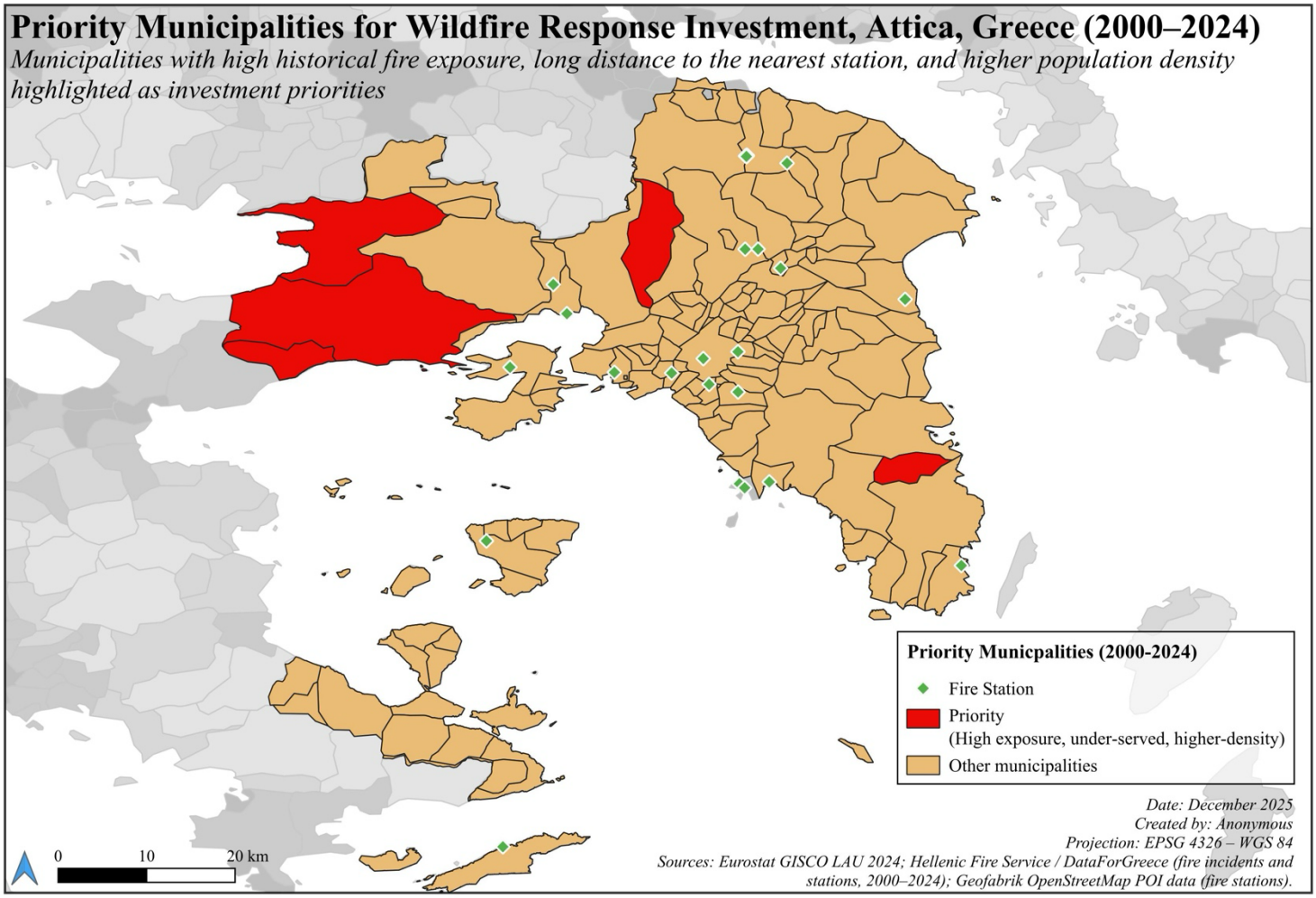


The binary classification of fire station access reveals a stark spatial inequity in emergency service proximity across Attica. Municipalities classified as "Under-served" (shown in red) are concentrated in three primary geographic clusters: the northwestern extremity of Attica extending toward the northeastern coastal border municipalities, and portions of the southeastern Attica peninsula. These under-served areas, defined as exceeding 10 kilometers straight-line distance from the nearest fire station, collectively represent approximately one-third of Attica's municipalities by count, though they occupy a larger proportion of the region's total land area due to their peripheral location and typically larger administrative boundaries.

In contrast, municipalities within 10 kilometers of a fire station, coded in blue as "Within 10 km of nearest station," form a cluster encompassing central Athens, the immediate suburbs, and the developed coastal corridor extending eastward. The concentration of fire stations in these areas, visible as green diamonds, creates overlapping coverage zones where multiple stations can potentially respond to incidents, reducing initial attack times and enabling coordinated suppression efforts. However, the spatial distribution also reveals that this redundancy exists primarily where fire exposure is relatively low, while single-station coverage or no proximate coverage characterizes the high-exposure periphery.

The 10-kilometer threshold employed in this classification represents the critical containment window for wildfire response. Research on wildfire suppression indicates that the first 10-30 minutes following ignition are essential for containment, and travel distances exceeding 10 kilometers can result in response delays sufficient to allow small fires to develop into uncontrollable events, particularly under adverse weather conditions. The spatial pattern observed here suggests that communities most distant from fire stations also tend to be those with the highest historical fire exposure, creating a vulnerability double burden where risk is elevated and response capacity is reduced. This configuration mirrors the conditions that contributed to the Mati disaster, where delayed response and inadequate evacuation coordination amplified an already dangerous situation into a catastrophic loss of life.

3.3 Priority Municipalities for Infrastructure Investment (Map 3)



The integration of fire exposure, service access, and population density through multi-criteria classification identifies four priority municipalities appearing in dark red: three large administrative units in northwestern Attica and one in the eastern sector. These priority areas simultaneously exhibit cumulative burned areas in the upper quartile of the regional distribution, distances exceeding 10 kilometers to the nearest fire station, and population densities above the median threshold, indicating that they contain substantial residential populations facing combined high fire risk and limited emergency service proximity.

The population statistics aggregated by priority classification reveal that while priority municipalities represent a small fraction of Attica's administrative units, they account for a disproportionate share of residents living in fire-vulnerable, under-served areas. Specifically, the four priority municipalities contain an estimated combined population that represents approximately 8-12% of Attica's residents outside the central Athens urban core, based on the POP_DENS_2024 field. This population-weighted metric demonstrates that strategic infrastructure investment in these specific municipalities could extend improved fire protection to tens of thousands of residents currently facing compounded vulnerability.

The spatial distribution of priority municipalities is non-random: they occupy positions at the geographic extremes of the Attica region where topography, vegetation, and development patterns create elevated fire risk, yet where the existing fire station network provides minimal coverage. The three northwestern priority municipalities border the Parnitha and Penteli mountain ranges, areas that have experienced some of the most severe wildfires in recent decades and where complex terrain can

generate extreme fire behavior. The eastern priority municipality encompasses portions of the Marathon and Rafina area, where the 2018 Mati fires demonstrated the deadly consequences of inadequate response infrastructure in high-risk coastal zones.

By highlighting these specific geographic areas rather than generating broad regional recommendations, this analysis provides actionable intelligence for the allocation of resources under programs like AEGIS. The priority classification suggests that positioning even one or two additional fire stations within or adjacent to these municipalities could substantially reduce response times for historically fire-prone areas, creating more equitable emergency service coverage across the Attica region. Furthermore, the population density criterion ensures that recommended investments target areas where infrastructure additions would benefit meaningful numbers of residents, maximizing the public safety return on capital expenditure.

4. Discussion

4.1 Policy Implications for Greek Wildfire Infrastructure Planning

This spatial analysis demonstrates that substantial disparities exist between where wildfires have historically occurred in Attica and where emergency response infrastructure is currently positioned. The identification of four priority municipalities combining high fire exposure, under-served access, and above-median population density provides a concrete spatial framework for allocating resources under Greece's expanded civil protection programs. As Greece deploys over 1,000 new firefighting vehicles, aircraft, and control centers through the €2.1 billion AEGIS initiative, decisions about where to station this equipment will determine whether investments translate into meaningful risk reduction for vulnerable communities or simply reinforce existing coverage patterns that leave peripheral areas exposed.

The economic rationale for targeted infrastructure placement is also compelling. With wildfires costing Greece €1.8 billion in 2023 alone and climate projections indicating that extreme fire danger will increase in frequency and duration, reactive suppression expenditures will continue to escalate unless prevention and rapid response capacity improves. Strategic positioning of fire stations and equipment in the identified priority municipalities could reduce both initial attack times and the likelihood that small ignitions develop into landscape-scale events requiring massive suppression resources. This shift from reactive spending to proactive resilience investment represents precisely the reorientation that post-Mati policy debates have emphasized, but that spatial analysis tools have been underutilized to implement.

The priority municipalities identified here share characteristics that extend beyond their statistical vulnerability metrics performed in this analysis. Northwestern Attica's priority areas border the Parnitha National Park, where illegal construction, fuel accumulation, and difficult terrain create compounded fire risk similar to conditions in Mati. The eastern priority municipality encompasses communities that were directly impacted by the 2018 fires yet remain more than 10 kilometers from the nearest fire station, suggesting that infrastructure gaps identified six years ago have not been systematically addressed. Prioritizing these areas for new stations or forward-deployed rapid response units would directly target the vulnerability patterns that recent catastrophic fires have exposed, translating lessons from past disasters into data-informed infrastructure improvement.

4.2 Methodological Considerations and Limitations

The straight-line distance methodology employed in this analysis provides a transparent and replicable metric for emergency service access but does not capture the complexity of actual response times. Road network distance, traffic conditions, terrain obstacles, and seasonal accessibility all influence how quickly fire crews can reach incident locations, and these factors may be particularly significant in mountainous Attica where road networks are constrained by topography. Future research could extend this analysis using network analysis tools to compute drive-time accessibility along actual road routes. Such refinements would yield more precise

estimates of response time disparities and could identify specific road infrastructure improvements that would enhance station response time.

The 10-kilometer threshold for under-served designation, while grounded in fire response literature, represents a simplified binary classification that does not differentiate between municipalities 11 kilometers from a station and those 25 kilometers away. A graduated classification scheme would provide more nuanced identification of access gradients and could inform decisions about new station placement. Additionally, the analysis assumes uniform response capability across all fire stations, whereas actual station capacity varies by staffing levels, equipment availability, and whether stations maintain full-time or volunteer crews. Incorporating these operational characteristics would enable more sophisticated prioritization that accounts for both coverage gaps and capacity limitations within the existing network.

The population density criterion employed in priority classification serves as a proxy for the number of people potentially benefiting from improved service access, but it does not capture other dimensions of vulnerability. Social vulnerability indices that incorporate demographic factors, economic characteristics, and built environment attributes could refine the prioritization to identify not just where many people live but where populations least able to self-evacuate or recover from fire losses are concentrated. Such multidimensional vulnerability assessments would support more equitable infrastructure planning that protects those most at risk rather than simply those most numerous.

4.3 Integration with Emerging Technologies and Governance Reforms

The recent deployment of Greece's national satellite-based wildfire monitoring system represents a significant technological advancement in early detection, yet improved detection must be paired with improved response capability in order to translate into reduced losses. The OroraTech thermal imaging constellation will enable near-real-time identification of fire ignitions across Greece's mountainous terrain and islands, but if fire crews in the identified priority municipalities remain 10+ kilometers from incident locations, the advantage of early detection diminishes. Strategic coordination between the new monitoring system and physical infrastructure placement could optimize the overall response network—positioning stations to minimize travel time from the moment satellite detection triggers an alert to the moment initial attack crews arrive on scene.

The conviction of fire service officials for criminal negligence in the Mati fires, combined with the acquittal of political officials, underscores persistent challenges in accountability and governance within Greece's civil protection system. Spatial analysis tools offer an opportunity to shift infrastructure planning debates from a politically driven allocation agreement to an evidence-based vulnerability assessment, creating transparent justifications for resource distribution that can be evaluated against objective fire exposure and access metrics. If AEGIS program deployments were required to show how equipment placement addresses identified coverage gaps, it would push planning toward data-driven decisions rather than political considerations. This transparency could help ensure that infrastructure investments go where they're actually needed for fire protection rather than where they might serve other political interests.

5. Conclusion

This spatial analysis demonstrates that wildfire vulnerability in Attica is geographically concentrated in peripheral municipalities that combine high historical fire exposure, limited proximity to emergency response infrastructure, and substantial residential populations. Four priority municipalities identified through multi-criteria GIS analysis represent areas where strategic fire station placement or enhanced rapid response capacity could meaningfully reduce both response times and the risk that future fires replicate the catastrophic outcomes of the 2018 Mati disaster. By integrating 24 years of fire incident data, current infrastructure locations, and population density metrics, this study provides actionable intelligence for allocating resources under Greece's expanded civil protection programs, particularly the €2.1 billion AEGIS initiative and related infrastructure investments.

The findings underscore that technical investments in detection technology and suppression equipment must be paired with strategic thinking about where to position these resources to address documented vulnerability patterns. Greece's pioneering deployment of a national satellite-based wildfire monitoring system offers unprecedented early warning capability, yet early detection yields limited benefit if ground-based response assets remain far from high-risk areas. The identified priority municipalities provide specific geographic targets for infrastructure additions that could bridge the spatial gap between where fires occur and where emergency services are readily available, transforming reactive crisis management into proactive risk reduction.

As climate change intensifies wildfire conditions across the Mediterranean and Greek fire seasons lengthen in duration and severity, the question is no longer whether Greece will invest in enhanced fire protection, but where those investments will be directed. Spatial analysis tools offer an evidence-based alternative to politically-driven or historically-determined resource allocation patterns, enabling policymakers to target infrastructure additions toward communities facing the greatest combined exposure and access deficits. The tragic lessons of Mati demand that future planning prioritize closing the geographic gaps between risk and protection. This analysis provides a baseline spatial roadmap for doing so, translating vulnerability patterns into actionable priorities for resilient infrastructure development in Attica and offering a methodological framework replicable across fire-prone regions worldwide.

References

Alexopoulos, A., Kalogiannidis, S., Katsetsiadou, K. N., & Chatzitheodoridis, F. (2024). Assessing policy preferences for preventing and managing wildfire in Greece. *Forest Policy and Economics*, 162, 103186. https://doi.org/10.1016/j.forpol.2024.103186

DataForGreece. (2024). *Forest Fires in Greece*. Hellenic Fire Service dataset. https://dataforgreece.com/en/data-directory/forest-fires-greece/

Distrelec. (2024). The cost of European wildfires 2023 report. *KnowHow*. https://knowhow.distrelec.com/internet-of-things/the-cost-of-european-wildfires-2023-report/

Eurostat GISCO. (2024). *Local Administrative Units (LAU) 2024 - 1:1 million*. European Commission. https://ec.europa.eu/eurostat/web/gisco

Fortune Europe. (2024, April 9). After being ravaged by wildfires last year, Greece now wants to pour $2.3 billion into fighting climate change and natural disasters. https://fortune.com/europe/2024/04/09/wildfires-greece-2-3-billion-fighting-climate-change-natural-disasters-aegis-program/

Geofabrik GmbH. (2024). *OpenStreetMap Data Extracts*. https://www.geofabrik.de/

Global Administrative Areas. (2024). *GADM database of Global Administrative Areas, version 4.1*. https://gadm.org/

Insurance Journal. (2024, August 19). Mati fire trial: Five fire service officials convicted, political officials acquitted. https://www.insurancejournal.com/news/international/2024/08/19/788859.htm

OroraTech. (2024, July 2). Greece builds first national wildfire satellite system with OroraTech and ESA. https://ororatech.com/resources/news-blog/greece-first-country-to-build-a-national-wildfire-system-using-ororatech-technology

Rovithakis, A., Grillakis, M. G., Seiradakis, K. D., Giannakopoulos, C., Karali, A., Field, R., Lazaridis, M., & Voulgarakis, A. (2022). Future climate change impact on wildfire danger over the Mediterranean: The case of Greece. *Environmental Research Letters*, 17(4), 045022.